

1. Data and CT-to-Video Pathway

LUNA 25/NLST
Test Cohort

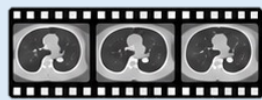
917 Annotations
318 Patients
836 Benign / 81 Malignant

Nodule-Centered
CT Volume



$64 \times 128 \times 128$
HU Clip [-1000, 1000]

CT-to-Video
Conversion



20 fps Grayscale MP4

2. Prompted VLM Experiment

CT-to-Video
Conversion



20 fps Grayscale MP4

VLM

Gemini 3 Flash Preview

Prompt Conditions (Z1-Z3/F1-F3)

	Minimal Prompt	Rich Prompt	Rich Prompt + Clinical Text
Zero-Shot	Z1	Z2	Z3
20-Shot	F1	F2	F3

Structured JSON Output

{“confidence”, “reasoning”}

3. Evaluation and Adaptation Spectrum

Weak / Non-Task-Trained

3D ResNet18 + TTT, 20-Shot
MedGemma under F3

Prompted
General-Purpose VLMs

Gemini Z/F Ablation
F0 Text-Only Control
Gemini Generations under F3

Full Task-Specific
Supervised 3D DL

7 Supervised 3D DL Baselines
Full LUNA25 Task Training

Validation and Comparators

LUNA25 Held-Out Test
Brock Clinical Model
LNDb External Validation